

Complex Analysis

with Applications

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a **complex number** z is an ordered pair (x, y) of real numbers x and y , written

$$z = (x, y).$$

x is called the **real part** and y the **imaginary part** of z , written

$$x = \operatorname{Re} z, \quad y = \operatorname{Im} z.$$

By definition, two complex numbers are **equal** if and only if their real parts are equal and their imaginary parts are equal.

$(0, 1)$ is called the **imaginary unit** and is denoted by i ,

(1)

$$i = (0, 1).$$

Addition of two complex numbers $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ is defined by

$$(2) \quad z_1 + z_2 = (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2).$$

Multiplication is defined by

$$(3) \quad z_1 z_2 = (x_1, y_1)(x_2, y_2) = (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1).$$

These two definitions imply that

$$(x_1, 0) + (x_2, 0) = (x_1 + x_2, 0)$$

and

$$(x_1, 0)(x_2, 0) = (x_1 x_2, 0)$$

as for real numbers x_1, x_2 . Hence the complex numbers "*extend*" the real numbers. We can thus write

$$(x, 0) = x. \quad \text{Similarly,} \quad (0, y) = iy$$

because by (1), and the definition of multiplication, we have

$$iy = (0, 1)y = (0, 1)(y, 0) = (0 \cdot y - 1 \cdot 0, 0 \cdot 0 + 1 \cdot y) = (0, y).$$

Together we have, by addition, $(x, y) = (x, 0) + (0, y) = x + iy$.

In practice, complex numbers $z = (x, y)$ are written

$$(4) \quad z = x + iy$$

or $\underline{z} = x + yi$, e.g., $17 + 4i$ (instead of $i4$).

If $x = 0$, then $z = iy$ and is called **pure imaginary**. Also, (1) and (3) give

$$(5) \quad i^2 = -1$$

because, by the definition of multiplication, $i^2 = i \cdot i = (0, 1)(0, 1) = (-1, 0) = -1$.

For **addition** the standard notation (4) gives see (2)

$$(x_1 + iy_1) + (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2).$$

For **multiplication** the standard notation gives the following very simple recipe. Multiply each term by each other term and use $i^2 = -1$ when it occurs see (3) :

$$\begin{aligned}(x_1 + iy_1)(x_2 + iy_2) &= x_1x_2 + ix_1y_2 + iy_1x_2 + i^2y_1y_2 \\ &= (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1).\end{aligned}$$

Subtraction, Division

Subtraction and **division** are defined as the inverse operations of addition and multiplication, respectively. Thus the **difference** $z = z_1 - z_2$ is the complex number z for which $z_1 = z + z_2$. Hence by (2),

$$(6) \quad z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2).$$

The **quotient** $z = z_1/z_2$ ($z_2 \neq 0$) is the complex number z for which $z_1 = zz_2$. If we equate the real and the imaginary parts on both sides of this equation, setting $z = x + iy$, we obtain $x_1 = x_2x - y_2y$, $y_1 = y_2x + x_2y$. The solution is

$$(7) \quad z = \frac{z_1}{z_2} = x + iy, \quad x = \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2}, \quad y = \frac{x_2y_1 - x_1y_2}{x_2^2 + y_2^2}.$$

The *practical rule* used to get this is by multiplying numerator and denominator of z_1/z_2 by $x_2 - iy_2$ and simplifying:

$$(7) \quad z = \frac{x_1 + iy_1}{x_2 + iy_2} = \frac{(x_1 + iy_1)(x_2 - iy_2)}{(x_2 + iy_2)(x_2 - iy_2)} = \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + i \frac{x_2y_1 - x_1y_2}{x_2^2 + y_2^2}.$$

The Imaginary Unit

The symbol i was originally used as a disguise for the embarrassing symbol $\sqrt{-1}$. We now say that i is the **imaginary unit** and define it by the property $i^2 = -1$. Using the imaginary unit, we build a general complex number out of two real numbers.

Definition Complex Number

A complex number is any number of the form $z = a + ib$ where a and b are real numbers and i is the imaginary unit.

The notations $a + ib$ and $a + bi$ are used interchangeably. The real number a in $z = a + ib$ is called the **real part** of z ; the real number b is called the **imaginary part** of z . The real and imaginary parts of a complex number z are abbreviated $\text{Re}(z)$ and $\text{Im}(z)$, respectively. For example, if $z = 4 - 9i$, then $\text{Re}(z) = 4$ and $\text{Im}(z) = -9$. A real constant multiple of the imaginary unit is called a **pure imaginary number**. For example, $z = 6i$ is a pure imaginary number. Two complex numbers are **equal** if their corresponding real and imaginary parts are equal.

Definition Equality

Complex numbers $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ are equal, $z_1 = z_2$, if $a_1 = a_2$ and $b_1 = b_2$.

In terms of the symbols $\text{Re}(z)$ and $\text{Im}(z)$, Definition states that $z_1 = z_2$ if $\text{Re}(z_1) = \text{Re}(z_2)$ and $\text{Im}(z_1) = \text{Im}(z_2)$.

The totality of complex numbers or the set of complex numbers is usually denoted by the symbol $\mathbf{C}$. Because *any* real number a can be written as $z = a + 0i$, we see that the set $\mathbf{R}$ of real numbers is a subset of $\mathbf{C}$.

Arithmetic Operations Complex numbers can be added, subtracted, multiplied, and divided. If $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$, these operations are defined as follows.

$$\text{Addition:} \quad z_1 + z_2 = (a_1 + ib_1) + (a_2 + ib_2) = (a_1 + a_2) + i(b_1 + b_2)$$

$$\text{Subtraction:} \quad z_1 - z_2 = (a_1 + ib_1) - (a_2 + ib_2) = (a_1 - a_2) + i(b_1 - b_2)$$

$$\begin{aligned} \text{Multiplication:} \quad z_1 \cdot z_2 &= (a_1 + ib_1)(a_2 + ib_2) \\ &= a_1a_2 - b_1b_2 + i(b_1a_2 + a_1b_2) \end{aligned}$$

$$\begin{aligned} \text{Division:} \quad \frac{z_1}{z_2} &= \frac{a_1 + ib_1}{a_2 + ib_2}, \quad a_2 \neq 0, \text{ or } b_2 \neq 0 \\ &= \frac{a_1a_2 + b_1b_2}{a_2^2 + b_2^2} + i \frac{b_1a_2 - a_1b_2}{a_2^2 + b_2^2} \end{aligned}$$

The familiar commutative, associative, and distributive laws hold for complex numbers:

$$\text{Commutative laws:} \quad \begin{cases} z_1 + z_2 = z_2 + z_1 \\ z_1 z_2 = z_2 z_1 \end{cases}$$

$$\text{Associative laws:} \quad \begin{cases} z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3 \\ z_1(z_2 z_3) = (z_1 z_2) z_3 \end{cases}$$

$$\text{Distributive law:} \quad z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$$

Addition, Subtraction, and Multiplication

- (i) To add (subtract) two complex numbers, simply add (subtract) the corresponding real and imaginary parts.
- (ii) To multiply two complex numbers, use the distributive law and the fact that $i^2 = -1$.

EXAMPLE 1 Addition and Multiplication

If $z_1 = 2 + 4i$ and $z_2 = -3 + 8i$, find (a) $z_1 + z_2$ and (b) $z_1 z_2$.

Solution (a) By adding real and imaginary parts, the sum of the two complex numbers z_1 and z_2 is

$$z_1 + z_2 = (2 + 4i) + (-3 + 8i) = (2 - 3) + (4 + 8)i = -1 + 12i.$$

(b) By the distributive law and $i^2 = -1$, the product of z_1 and z_2 is

$$\begin{aligned} z_1 z_2 &= (2 + 4i)(-3 + 8i) = (2 + 4i)(-3) + (2 + 4i)(8i) \\ &= -6 - 12i + 16i + 32i^2 \\ &= (-6 - 32) + (16 - 12)i = -38 + 4i. \end{aligned}$$

Zero and Unity The zero in the complex number system is the number $0 + 0i$ and the **unity** is $1 + 0i$. The zero and unity are denoted by 0 and 1, respectively. The zero is the **additive identity** in the complex number system since, for any complex number $z = a + ib$, we have $z + 0 = z$. To see this, we use the definition of addition:

$$z + 0 = (a + ib) + (0 + 0i) = a + 0 + i(b + 0) = a + ib = z.$$

Similarly, the unity is the **multiplicative identity** of the system since, for any complex number z , we have $z \cdot 1 = z \cdot (1 + 0i) = z$.

Conjugate If z is a complex number, the number obtained by changing the sign of its imaginary part is called the **complex conjugate**, or simply **conjugate**, of z and is denoted by the symbol $\bar{z}$. In other words, if $z = a + ib$, then its conjugate is $\bar{z} = a - ib$. For example, if $z = 6 + 3i$, then $\bar{z} = 6 - 3i$; if $z = -5 - i$, then $\bar{z} = -5 + i$. If z is a real number, say, $z = 7$, then $\bar{z} = 7$.

From the definitions of addition and subtraction of complex numbers, it is readily shown that the conjugate of a sum and difference of two complex numbers is the sum and difference of the conjugates:

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2, \quad \overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2. \quad (1)$$

Moreover, we have the following three additional properties:

$$\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2, \quad \overline{\left(\frac{z_1}{z_2} \right)} = \frac{\bar{z}_1}{\bar{z}_2}, \quad \bar{\bar{z}} = z. \quad (2)$$

the conjugate of any finite sum (product) of complex numbers is the sum (product) of the conjugates.

The definitions of addition and multiplication show that the sum and product of a complex number z with its conjugate $\bar{z}$ is a real number:

$$z + \bar{z} = (a + ib) + (a - ib) = 2a \quad (3)$$

$$z\bar{z} = (a + ib)(a - ib) = a^2 - i^2b^2 = a^2 + b^2. \quad (4)$$

The difference of a complex number z with its conjugate $\bar{z}$ is a pure imaginary number:

$$z - \bar{z} = (a + ib) - (a - ib) = 2ib. \quad (5)$$

Since $a = \operatorname{Re}(z)$ and $b = \operatorname{Im}(z)$, (3) and (5) yield two useful formulas:

$$\operatorname{Re}(z) = \frac{z + \bar{z}}{2} \quad \text{and} \quad \operatorname{Im}(z) = \frac{z - \bar{z}}{2i}. \quad (6)$$

$$\overline{(z_1 + z_2)} = \bar{z}_1 + \bar{z}_2, \quad \overline{(z_1 - z_2)} = \bar{z}_1 - \bar{z}_2,$$

$$\overline{(z_1 z_2)} = \bar{z}_1 \bar{z}_2, \quad \overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}.$$

PROVE THE EQUALITIES

Let $z_1 = 4 + 3i$ and $z_2 = 2 + 5i$.

$$\operatorname{Im} z_1 = \frac{1}{2i} (4 + 3i) - (4 - 3i) = \frac{3i + 3i}{2i} = 3.$$

Also, the multiplication formula is verified by

$$\overline{(z_1 z_2)} = \overline{(4 + 3i)(2 + 5i)} = \overline{(-7 + 26i)} = -7 - 26i,$$

$$\bar{z}_1 \bar{z}_2 = (4 - 3i)(2 - 5i) = -7 - 26i.$$

Division

To divide z_1 by z_2 , multiply the numerator and denominator of z_1/z_2 by the conjugate of z_2 . That is,

$$\frac{z_1}{z_2} = \frac{z_1}{z_2} \cdot \frac{\bar{z}_2}{\bar{z}_2} = \frac{z_1 \bar{z}_2}{z_2 \bar{z}_2} \quad (7)$$

and then use the fact that $z_2 \bar{z}_2$ is the sum of the squares of the real and imaginary parts of z_2 .

EXAMPLE 2 Division

If $z_1 = 2 - 3i$ and $z_2 = 4 + 6i$, find z_1/z_2 .

We multiply numerator and denominator by the conjugate $\bar{z}_2 = 4 - 6i$ of the denominator $z_2 = 4 + 6i$ and then use (4):

$$\frac{z_1}{z_2} = \frac{2 - 3i}{4 + 6i} = \frac{2 - 3i}{4 + 6i} \frac{4 - 6i}{4 - 6i} = \frac{8 - 12i - 12i + 18i^2}{4^2 + 6^2} = \frac{-10 - 24i}{52}.$$

Because we want an answer in the form $a + bi$, we rewrite the last result by dividing the real and imaginary parts of the numerator $-10 - 24i$ by 52 and reducing to lowest terms:

$$\frac{z_1}{z_2} = -\frac{10}{52} - \frac{24}{52}i = -\frac{5}{26} - \frac{6}{13}i.$$

Inverses In the complex number system, every number z has a unique **additive inverse**. As in the real number system, the additive inverse of $z = a + ib$ is its *negative*, $-z$, where $-z = -a - ib$. For any complex number z , we have $z + (-z) = 0$. Similarly, every *nonzero* complex number z has a **multiplicative inverse**. In symbols, for $z \neq 0$ there exists one and only one nonzero complex number z^{-1} such that $zz^{-1} = 1$. The multiplicative inverse z^{-1} is the same as the reciprocal $1/z$.

EXAMPLE 3 Reciprocal

Find the reciprocal of $z = 2 - 3i$.

By the definition of division we obtain

$$\frac{1}{z} = \frac{1}{2 - 3i} = \frac{1}{2 - 3i} \frac{2 + 3i}{2 + 3i} = \frac{2 + 3i}{4 + 9} = \frac{2 + 3i}{13}.$$

That is,
$$\frac{1}{z} = z^{-1} = \frac{2}{13} + \frac{3}{13}i.$$

You should take a few seconds to verify the multiplication

$$zz^{-1} = (2 - 3i) \left(\frac{2}{13} + \frac{3}{13}i \right) = 1.$$

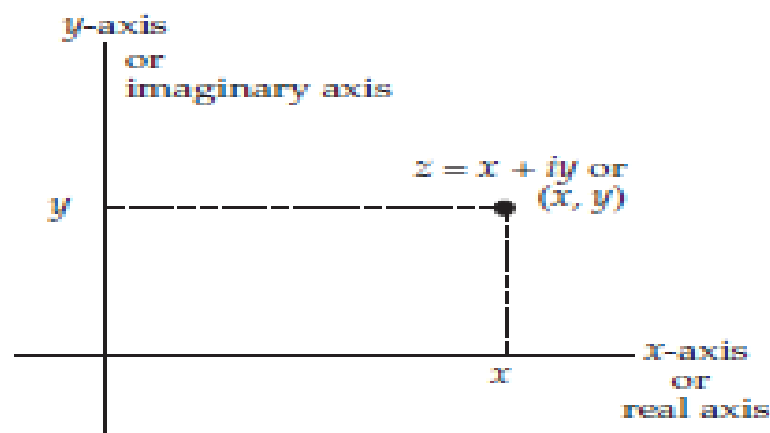
the concept of *order* in the real number system does not carry over to the complex number system. In other words, we cannot compare two complex numbers $z_1 = a_1 + ib_1$, $b_1 \neq 0$, and $z_2 = a_2 + ib_2$, $b_2 \neq 0$, by means of inequalities. Statements such as $z_1 < z_2$ or $z_2 \geq z_1$ have no meaning in $\mathbf{C}$ except in the special case when the two numbers z_1 and z_2 are real.

1.2 Complex Plane

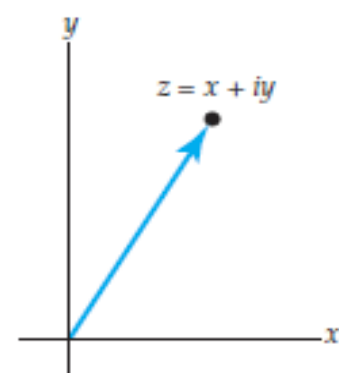
A complex number $z = x + iy$ is uniquely determined by an *ordered pair* of real numbers (x, y) . The first and second entries of the ordered pairs correspond, in turn, with the real and imaginary parts of the complex number. For example, the ordered pair $(2, -3)$ corresponds to the complex number $z = 2 - 3i$. Conversely, $z = 2 - 3i$ determines the ordered pair $(2, -3)$. The numbers 7 , i , and $-5i$ are equivalent to $(7, 0)$, $(0, 1)$, $(0, -5)$, respectively. In this manner we are able to associate a complex number $z = x + iy$ with a *point* (x, y) in a coordinate plane.

Complex Plane

Because of the correspondence between a complex number $z = x + iy$ and one and only one point (x, y) in a coordinate plane, we shall use the terms *complex number* and *point* interchangeably. The coordinate plane illustrated in Figure is called the **complex plane** or simply the **z -plane**. The horizontal or x -axis is called the **real axis** because each point on that axis represents a real number. The vertical or y -axis is called the **imaginary axis** because a point on that axis represents a pure imaginary number.



the numbers in an ordered pair of real numbers can be interpreted as the components of a vector. Thus, a complex number $z = x + iy$ can also be viewed as a two-dimensional position vector, that is, a vector whose initial point is the origin and whose terminal point is the point (x, y) . This vector interpretation prompts us to define the length of the vector z as the distance $\sqrt{x^2 + y^2}$ from the origin to the point (x, y) . This length is given a special name.



z as a position vector

The modulus of a complex number $z = x + iy$, is the real number

$$|z| = \sqrt{x^2 + y^2}. \quad (1)$$

The modulus $|z|$ of a complex number z is also called the absolute value of z . We shall use both words *modulus* and *absolute value*

EXAMPLE 1 Modulus of a Complex Number

If $z = 2 - 3i$, then from (1) we find the modulus of the number to be $|z| = \sqrt{2^2 + (-3)^2} = \sqrt{13}$. If $z = -9i$, then (1) gives $|-9i| = \sqrt{(-9)^2} = 9$.

Properties

for any complex number

$z = x + iy$ the product $z\bar{z}$ is a real number; specifically, $z\bar{z}$ is the sum of the squares of the real and imaginary parts of z : $z\bar{z} = x^2 + y^2$. Inspection of (1) then shows that $|z|^2 = x^2 + y^2$. The relations

$$|z|^2 = z\bar{z} \quad \text{and} \quad |z| = \sqrt{z\bar{z}} \quad (2)$$

deserve to be stored in memory. The modulus of a complex number z has the additional properties.

$$|z_1 z_2| = |z_1| |z_2| \quad \text{and} \quad \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}. \quad (3)$$

Note that when $z_1 = z_2 = z$, the first property in (3) shows that

$$|z^2| = |z|^2. \quad (4)$$

$$|z| \geq |\operatorname{Re}(z)| \geq \operatorname{Re}(z)$$

$$|z| \geq |\operatorname{Im}(z)| \geq \operatorname{Im}(z)$$

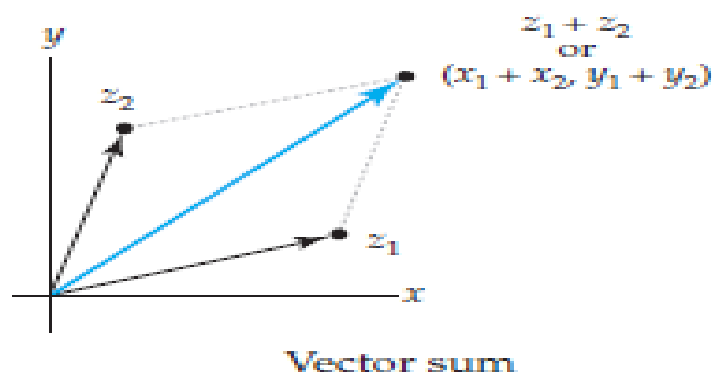
$$|z| = |\bar{z}|$$

Distance Again

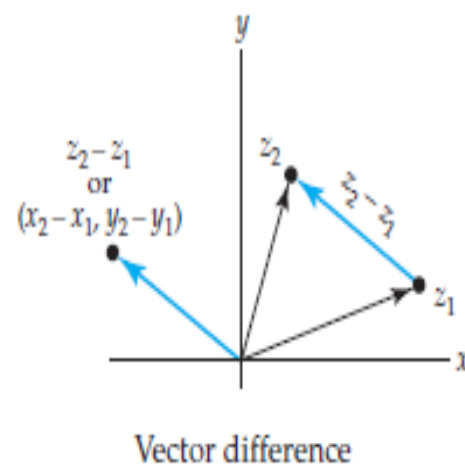
The addition of complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ when stated in terms of ordered pairs:

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$$

is simply the component definition of vector addition. The vector interpretation of the sum $z_1 + z_2$ is the vector shown in Figure 1 as the main diagonal of a parallelogram whose initial point is the origin and terminal point is $(x_1 + x_2, y_1 + y_2)$.



The difference $z_2 - z_1$ can be drawn either starting from the terminal point of z_1 and ending at the terminal point of z_2 , or as a position vector whose initial point is the origin and terminal point is $(x_2 - x_1, y_2 - y_1)$.



the distance between two points $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ in the complex plane is the same as the distance between the origin and the point $(x_2 - x_1, y_2 - y_1)$; that is, $|z| = |z_2 - z_1| = |(x_2 - x_1) + i(y_2 - y_1)|$ or

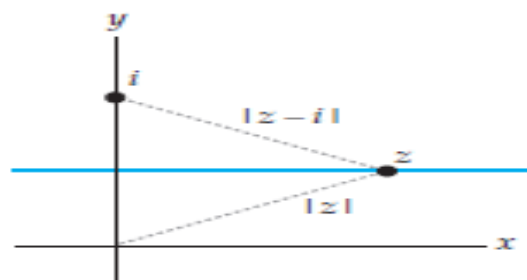
$$|z_2 - z_1| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}. \quad (5)$$

When $z_1 = 0$, we see again that the modulus $|z_2|$ represents the distance between the origin and the point z_2 .

EXAMPLE 2 Set of Points in the Complex Plane

Describe the set of points z in the complex plane that satisfy $|z| = |z - i|$.

We can interpret the given equation as equality of distances: The distance from a point z to the origin equals the distance from z to the point i .



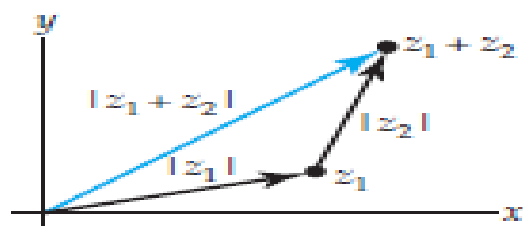
Horizontal line is the set of points satisfying $|z| = |z - i|$.

$$\begin{aligned}\sqrt{x^2 + y^2} &= \sqrt{x^2 + (y - 1)^2} \\ x^2 + y^2 &= x^2 + (y - 1)^2 \\ x^2 + y^2 &= x^2 + y^2 - 2y + 1.\end{aligned}$$

The last equation yields $y = \frac{1}{2}$. Since the equality is true for arbitrary x , $y = \frac{1}{2}$ is an equation of the horizontal line shown in color in Figure [1](#). Complex numbers satisfying $|z| = |z - i|$ can then be written as $z = x + \frac{1}{2}i$.

we pointed out that

no order relation can be defined on the system of complex numbers. However, since $|z|$ is a real number, we can compare the absolute values of two complex numbers. For example, if $z_1 = 3 + 4i$ and $z_2 = 5 - i$, then $|z_1| = \sqrt{25} = 5$ and $|z_2| = \sqrt{26}$ and, consequently, $|z_1| < |z_2|$. In view of (1), a geometric interpretation of the last inequality is simple: *The point (3, 4) is closer to the origin than the point (5, -1).*



Triangle with vector sides

$$|z_1 + z_2| \leq |z_1| + |z_2|. \quad \text{PROVE} \quad (6)$$

The result in (6) is known as the **triangle inequality**. Now from the identity $z_1 = z_1 + z_2 + (-z_2)$, (6) gives

$$|z_1| = |z_1 + z_2 + (-z_2)| \leq |z_1 + z_2| + |-z_2|.$$

Since $|z_2| = |-z_2|$

$$|z_1 + z_2| \geq |z_1| - |z_2|. \quad (7)$$

But because $z_1 + z_2 = z_2 + z_1$, (7) can be written in the alternative form $|z_1 + z_2| = |z_2 + z_1| \geq |z_2| - |z_1| = -(|z_1| - |z_2|)$ and so combined with the last result implies

$$|z_1 + z_2| \geq \big| |z_1| - |z_2| \big|. \quad (8)$$

It also follows from (6) by replacing z_2 by $-z_2$ that $|z_1 + (-z_2)| \leq |z_1| + |(-z_2)| = |z_1| + |z_2|$. This result is the same as

$$|z_1 - z_2| \leq |z_1| + |z_2|. \quad (9)$$

From (8) with z_2 replaced by $-z_2$, we also find

$$|z_1 - z_2| \geq \big| |z_1| - |z_2| \big|. \quad (10)$$

In conclusion, we note that the triangle inequality (6) extends to any finite sum of complex numbers:

$$|z_1 + z_2 + z_3 + \cdots + z_n| \leq |z_1| + |z_2| + |z_3| + \cdots + |z_n|. \quad (11)$$

EXAMPLE 3 An Upper Bound

Find an upper bound for $\left| \frac{-1}{z^4 - 5z + 1} \right|$ if $|z| = 2$.

Solution By the second result in (3), the absolute value of a quotient is the quotient of the absolute values. Thus with $|-1| = 1$, we want to find a positive real number M such that

$$\frac{1}{|z^4 - 5z + 1|} \leq M.$$

To accomplish this task we want the denominator as small as possible. By (10) we can write

$$|z^4 - 5z + 1| = |z^4 - (5z - 1)| \geq ||z^4| - |5z - 1||. \quad (12)$$

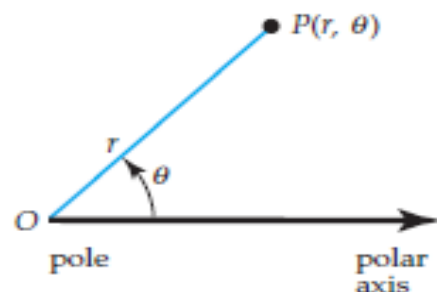
But to make the difference in the last expression in (12) as small as possible, we want to make $|5z - 1|$ as large as possible. From (9), $|5z - 1| \leq |5z| + |-1| = 5|z| + 1$. Using $|z| = 2$, (12) becomes

$$\begin{aligned} |z^4 - 5z + 1| &\geq ||z^4| - |5z - 1|| \geq \left| |z|^4 - (5|z| + 1) \right| \\ &= \left| |z|^4 - 5|z| - 1 \right| = |16 - 10 - 1| = 5. \end{aligned}$$

Hence for $|z| = 2$ we have

$$\frac{1}{|z^4 - 5z + 1|} \leq \frac{1}{5}.$$

Polar Form of Complex Numbers



Polar coordinates

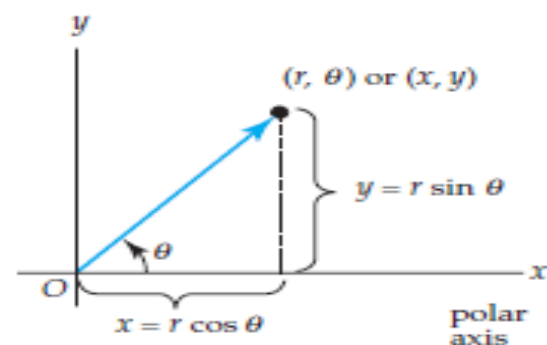
$$x = r \cos \theta, y = r \sin \theta.$$

$$z = x + iy \quad z = (r \cos \theta) + i(r \sin \theta)$$

$$z = r (\cos \theta + i \sin \theta). \quad (1)$$

We say that (1) is the polar form or polar representation of the complex number z . Again, from Figure we see that the coordinate r can be interpreted as the distance from the origin to the point (x, y) .

$$r = |z|.$$



Polar coordinates in the complex plane

The angle θ of inclination of the vector z , which *will always be measured in radians* from the positive real axis, is positive when measured counterclockwise and negative when measured clockwise. The angle θ is called an **argument** of z and is denoted by $\theta = \arg(z)$. An argument θ of a complex number must satisfy the equations $\cos \theta = x/r$ and $\sin \theta = y/r$. An argument of a complex number z is not unique since $\cos \theta$ and $\sin \theta$ are 2π -periodic; in other words, if θ_0 is an argument of z , then necessarily the angles $\theta_0 \pm 2\pi$, $\theta_0 \pm 4\pi, \dots$ are also arguments of z . In practice we use $\tan \theta = y/x$ to find θ . However, because $\tan \theta$ is π -periodic, some care must be exercised in using the last equation. A calculator will give only angles satisfying $-\pi/2 < \tan^{-1}(y/x) < \pi/2$, that is, angles in the first and fourth quadrants. We have to choose θ consistent with the quadrant in which z is located; this may require adding or subtracting π to $\tan^{-1}(y/x)$ when appropriate. The following example illustrates how this is done.

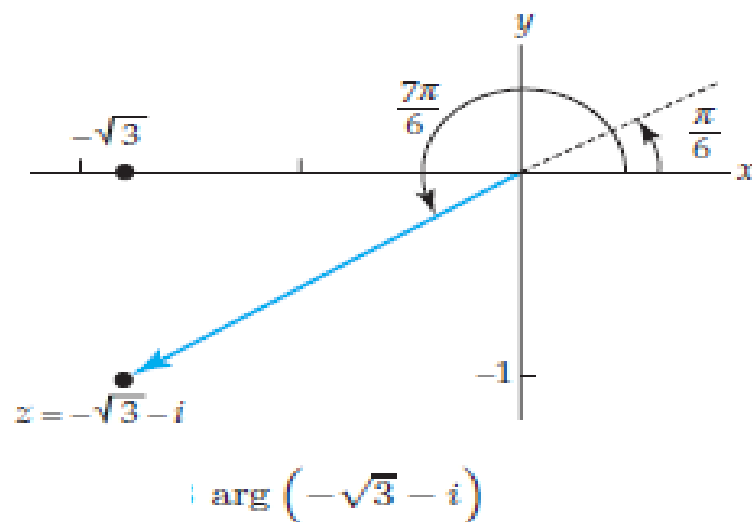
EXAMPLE 1 A Complex Number in Polar Form

Express $-\sqrt{3} - i$ in polar form.

Solution With $x = -\sqrt{3}$ and $y = -1$ we obtain $r = |z| = \sqrt{(-\sqrt{3})^2 + (-1)^2} = 2$. Now $y/x = -1/(-\sqrt{3}) = 1/\sqrt{3}$, and so a calculator gives $\tan^{-1}(1/\sqrt{3}) = \pi/6$, which is an angle whose terminal side is in the first quadrant. But since the point $(-\sqrt{3}, -1)$ lies in the third quadrant, we take the solution of $\tan \theta = -1/(-\sqrt{3}) = 1/\sqrt{3}$ to be $\theta = \arg(z) = \pi/6 + \pi = 7\pi/6$.

It follows from (1) that a polar form of the number is

$$z = 2 \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right). \quad (2)$$

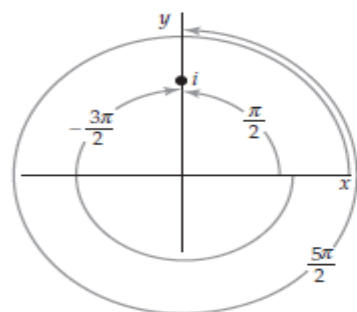


Principal Argument The symbol $\arg(z)$ actually represents a set of values, but the argument θ of a complex number that lies in the interval $-\pi < \theta \leq \pi$ is called the **principal value** of $\arg(z)$ or the **principal argument** of z . The principal argument of z is *unique* and is represented by the symbol $\text{Arg}(z)$, that is,

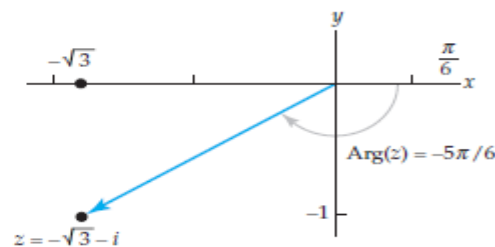
$$-\pi < \text{Arg}(z) \leq \pi.$$

For example, if $z = i$, some values of $\arg(i)$ are $\pi/2$, $5\pi/2$, $-3\pi/2$, and so on, but $\text{Arg}(i) = \pi/2$. Similarly, that the argument of $-\sqrt{3} - i$ that lies in the interval $(-\pi, \pi)$, the principal argument of z , is $\text{Arg}(z) = \pi/6 - \pi = -5\pi/6$. Using $\text{Arg}(z)$ we can express the complex number in (2) in the alternative polar form:

$$z = 2 \left[\cos \left(-\frac{5\pi}{6} \right) + i \sin \left(-\frac{5\pi}{6} \right) \right].$$



Some arguments of i



Principal argument

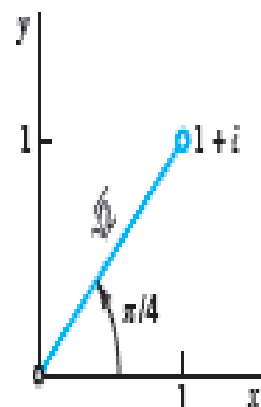
$$z = -\sqrt{3} - i$$

In general, $\arg(z)$ and $\text{Arg}(z)$ are related by

$$\arg(z) = \text{Arg}(z) + 2n\pi, \quad n = 0, \pm 1, \pm 2, \dots \quad (3)$$

For example, $\arg(i) = \frac{\pi}{2} + 2n\pi$. For the choices $n = 0$ and $n = -1$, (3) gives $\arg(i) = \text{Arg}(i) = \pi/2$ and $\arg(i) = -3\pi/2$, respectively.

Polar Form of Complex Numbers. Principal Value $\text{Arg } z$



$z = 1 + i$ has the polar form $z = \sqrt{2} (\cos \frac{1}{4}\pi + i \sin \frac{1}{4}\pi)$. Hence we obtain

$$|z| = \sqrt{2}, \quad \arg z = \frac{1}{4}\pi \pm 2n\pi \quad (n = 0, 1, \dots), \quad \text{and} \quad \text{Arg } z = \frac{1}{4}\pi \quad (\text{the principal value}).$$

Similarly, $z = 3 + 3\sqrt{3}i = 6 (\cos \frac{1}{3}\pi + i \sin \frac{1}{3}\pi)$, $|z| = 6$, and $\text{Arg } z = \frac{1}{3}\pi$.

CAUTION!

, we must pay attention to the quadrant in which z lies, since $\tan \theta$ has period π , so that the arguments of z and $-z$ have the same tangent. *Example:* for $\theta_1 = \arg (1 + i)$ and $\theta_2 = \arg (-1 - i)$ we have $\tan \theta_1 = \tan \theta_2 = 1$.

Multiplication and Division

The polar form of a complex number is especially convenient when multiplying or dividing two complex numbers. Suppose

$$z_1 = r_1(\cos \theta_1 + i \sin \theta_1) \quad \text{and} \quad z_2 = r_2(\cos \theta_2 + i \sin \theta_2),$$

where θ_1 and θ_2 are any arguments of z_1 and z_2 , respectively. Then

$$z_1 z_2 = r_1 r_2 [\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 + i (\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)] \quad (4)$$

and, for $z_2 \neq 0$,

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 + i (\sin \theta_1 \cos \theta_2 - \cos \theta_1 \sin \theta_2)]. \quad (5)$$

From the addition formulas[†] for the cosine and sine, (4) and (5) can be rewritten as

$$z_1 z_2 = r_1 r_2 [\cos (\theta_1 + \theta_2) + i \sin (\theta_1 + \theta_2)] \quad (6)$$

and

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos (\theta_1 - \theta_2) + i \sin (\theta_1 - \theta_2)]. \quad (7)$$

$$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2) \quad \text{and} \quad \arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2). \quad (8)$$

Proposition: Let $z_1 \neq 0$ and $z_2 \neq 0$

$$i) \arg \bar{z}_1 = -\arg z_1 \pmod{2\pi}$$

$\bar{z}_1 \cdot z_1$ is a real number. So,

$$\arg(z_1 \cdot \bar{z}_1) = 0 \pmod{2\pi}$$

$$\arg(z_1) + \arg(\bar{z}_1) = 0 \pmod{2\pi}$$

$$\arg(\bar{z}_1) = -\arg(z_1) \pmod{2\pi}$$

$$ii) \arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2 \pmod{2\pi}$$

$$\left(\frac{z_1}{z_2}\right) \cdot z_2 = z_1 \quad \text{is true.}$$

$$\arg\left(\left(\frac{z_1}{z_2}\right) \cdot z_2\right) = \arg z_1 \pmod{2\pi}$$

$$\arg\left(\frac{z_1}{z_2}\right) + \arg(z_2) = \arg z_1 \pmod{2\pi}$$

$$\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2 \pmod{2\pi}$$

$$\text{iii) } \arg(1/z_1) = -\arg z_1 \pmod{2\pi}$$

$$(1/z_1) \cdot z_1 = 1$$

$$\arg[(1/z_1) \cdot z_1] = \arg 1 \pmod{2\pi}$$

$$\arg(1/z_1) + \arg(z_1) = 0 \pmod{2\pi}$$

$$\arg(1/z_1) = -\arg(z_1)$$

EXAMPLE 2 Argument of a Product and of a Quotient

We have just seen that for $z_1 = i$ and $z_2 = -\sqrt{3} - i$ that $\text{Arg}(z_1) = \pi/2$ and $\text{Arg}(z_2) = -5\pi/6$, respectively. Thus arguments for the product and quotient

$$z_1 z_2 = i(-\sqrt{3} - i) = 1 - \sqrt{3}i \quad \text{and} \quad \frac{z_1}{z_2} = \frac{i}{-\sqrt{3} - i} = -\frac{1}{4} - \frac{\sqrt{3}}{4}i$$

can be obtained from (8):

$$\arg(z_1 z_2) = \frac{\pi}{2} + \left(-\frac{5\pi}{6}\right) = -\frac{\pi}{3} \quad \text{and} \quad \arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2} - \left(-\frac{5\pi}{6}\right) = \frac{4\pi}{3}.$$

Integer Powers of z We can find integer powers of a complex number z from the results in (6) and (7). For example, if $z = r(\cos \theta + i \sin \theta)$, then with $z_1 = z_2 = z$, (6) gives

$$z^2 = r^2 [\cos (\theta + \theta) + i \sin (\theta + \theta)] = r^2 (\cos 2\theta + i \sin 2\theta) .$$

Since $z^3 = z^2 z$, it then follows that

$$z^3 = r^3 (\cos 3\theta + i \sin 3\theta) ,$$

and so on. In addition, if we take $\arg(1) = 0$, then (7) gives

$$\frac{1}{z^2} = z^{-2} = r^{-2} [\cos(-2\theta) + i \sin(-2\theta)] .$$

Continuing in this manner, we obtain a formula for the n th power of z for any integer n :

$$z^n = r^n (\cos n\theta + i \sin n\theta) . \tag{9}$$

When $n = 0$, we get the familiar result $z^0 = 1$.

EXAMPLE 3 Power of a Complex Number

Compute z^3 for $z = -\sqrt{3} - i$.

Solution In (2) of Example 1 we saw that a polar form of the given number is $z = 2[\cos(7\pi/6) + i \sin(7\pi/6)]$. Using (9) with $r = 2$, $\theta = 7\pi/6$, and $n = 3$ we get

$$\left(-\sqrt{3} - i\right)^3 = 2^3 \left[\cos\left(3\frac{7\pi}{6}\right) + i \sin\left(3\frac{7\pi}{6}\right) \right] = 8 \left[\cos \frac{7\pi}{2} + i \sin \frac{7\pi}{2} \right] = -8i$$

since $\cos(7\pi/2) = 0$ and $\sin(7\pi/2) = -1$.

de Moivre's Formula

When $z = \cos \theta + i \sin \theta$, we have $|z| = r = 1$, and so (9) yields

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta. \quad (10)$$

This last result is known as **de Moivre's formula**

EXAMPLE 4 de Moivre's Formula

From (10), with $\theta = \pi/6$, $\cos \theta = \sqrt{3}/2$ and $\sin \theta = 1/2$:

$$\begin{aligned}\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^3 &= \cos 3\theta + i \sin 3\theta = \cos\left(3 \cdot \frac{\pi}{6}\right) + i \sin\left(3 \cdot \frac{\pi}{6}\right) \\ &= \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = i.\end{aligned}$$

show the following inequality

Ex: $\sqrt{2} |z| \geq |\operatorname{Re} z| + |\operatorname{Im} z|$

$x = \operatorname{Re} z$, $y = \operatorname{Im} z$

$$(|x| - |y|)^2 \geq 0$$

$$|x|^2 - 2|x||y| + |y|^2 \geq 0$$

$$|x|^2 + |y|^2 \geq 2|x||y| \quad (\text{add } |x|^2 + |y|^2 \text{ to the both side})$$

$$2(|x|^2 + |y|^2) \geq x^2 + 2|x||y| + |y|^2$$

$$2|z|^2 \geq (|x| + |y|)^2$$

$$\sqrt{2} |z| \geq |x| + |y| = |\operatorname{Re} z| + |\operatorname{Im} z|$$

Ex: if $|z| = 1$; $|z^2 + z + 1| \leq |z|^2 + |z| + 1 = 3$

$$|z^3 - 2| \geq ||z|^3 - 2| = |1 - 2| = 1.$$